

HealthConnect Clinic, Week 7

Analytics Testing & Refinement Report

Data Analytics Track: Testing, Refinement & End-to-End Validation

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1. Week 6 → Week 7 Transition

Main Week 6 output: an updated Power BI dashboard with bootstrap confidence-interval error bars on all four KPI rate charts, plus a supporting notebook reproducing the bootstrap validation with real outputs, and a written report deepening and validating the Week 5 findings.

Most important Week 6 result: the booking lead-time finding was confirmed as the strongest, most validated result in the project, by Week 5 point estimates, Week 6 bootstrap confidence intervals, and independently by the DS track collaborator's cross-validated model, where it is the dominant predictive feature.

Main issue/limitation carried forward from Week 6: the distance "15km cliff" finding was downgraded. Bootstrap CIs showed the three middle distance bands are statistically indistinguishable from each other, confirmed independently by the DS track collaborator's permutation importance check. This was documented in the Week 6 text, but the dashboard chart itself had not yet been updated to reflect the downgrade. Week 6 also left two questions open: whether distance overlaps with lead time (which would explain its low model weight), and how a double-risk intervention pilot would be run.

Component tested this week: the five core KPI values (KPIs 1 to 4 with error bars, plus the KPI 5 cancellation-to-no-show ratio chart), the lead-time and distance-band findings, the double-risk segment definition, and the DS track collaborator's `double_risk_flag` feature that is meant to encode the double-risk finding.

Track(s) relevant: Data Analytics (primary), DS track collaborator (cross-track validation).

What was tested and what was expected to improve/validate: whether the displayed KPI values are correct and reproducible, whether the lead-time and distance findings hold up under further scrutiny (including whether distance is simply redundant with lead time), and whether the DS track collaborator's model correctly encodes the DA-validated double-risk finding it is documented as being built on.

2. Testing Approach & Objectives

The Week 7 testing objective was to determine four things:

- Are the KPI values accurate and reproducible from the underlying data, independent of the notebook that originally produced them? This covers all five KPIs, including the cancellation-to-no-show ratio (bottom-right chart on the dashboard, no error bars).
- Do the two headline findings (lead-time risk, distance-band risk) hold up when tested against segments and checks not previously run, including whether the distance effect is redundant with lead time?
- Does the DS track collaborator's feature engineering correctly reflect the Data Analytics findings it is built on, under both the Random Forest and the exact Week 6 candidate model (a Logistic Regression)?
- How could the double-risk finding be turned into a real intervention pilot? (Design only this week.)

Testing evidence for each item is captured in the test table (Section 3) and the cross-track record (Section 6), all computed directly from the shared prepared dataset (`healthconnect_appointments_prepared.csv`) and the Week 6 KPI confidence-interval tables. The notebook and two scripts in the repository reproduce every number.

3. Test Table: KPI Validation, Segment Testing & Double-Risk Investigation

Component Tested	Test / Scenario	Expected Result	Actual Result	Pass/Fail	Action / Retest
KPI 1: Prior no-show bracket	Dashboard % vs source CSV	Exact match	43.5/53.5/59.4/68.8%, exact match	PASS	None needed
KPI 2: Lead-time band	Dashboard % vs source CSV	Exact match	27.8/33.6/43.2/53.8/67.7%, exact match	PASS	None needed
KPI 3: Distance band	Dashboard % vs source CSV	Exact match	46.5/46.5/48.5/54.1%, exact match	PASS	None needed
KPI 4: Reminder sent	Dashboard % vs source CSV	Exact match	47.4% / 51.4%, exact match	PASS	None needed
All 4 dashboard KPIs	Independent recompute directly from raw prepared CSV, bypassing the notebook entirely	Reproduce same values	Matched to 4 decimal places on all four	PASS	Confirms pipeline is reproducible, not a notebook-only artefact
KPI 5: Cancellations vs no-shows ratio (new)	Recompute from prepared CSV and compare with the Week 6 KPI Validation report (the dashboard's KPI 5 chart, bottom right, has no error bars)	Reproduce Week 6 point estimates	Overall 0.109; by lead band 0.202 / 0.139 / 0.118 / 0.105 / 0.076, all matching Week 6 to 3 decimals. Fresh bootstrap CIs within 0.007 of Week 6.	PASS	Closes the coverage gap: all five KPIs now tested. The chart's bars still need to be read in Power BI against these values.
Lead-time finding	Segment test against appointment_type (new segment)	Direction holds in every subgroup	46–60 day band highest no-show rate in all 4 appointment types	PASS	None needed
Lead-time finding	Segment test against age_group (new segment)	Direction holds in every subgroup	46–60 day band highest in all 6 age groups	PASS	None needed
Distance band finding	CI overlap check on the 3 middle bands	CIs should overlap, confirming the Week 6 downgrade	All 3 consecutive-pair CIs overlap	PASS	Text finding confirmed; chart still needs a visual fix (see Section 5)
Distance vs lead time (new)	Correlation, and 15+ km vs under-15 km gap inside each lead-time band (4,910 rows with known distance)	If distance were redundant with lead time, the gap would vanish inside bands	Correlation –0.008. 15+ km is higher in all 5 bands (+3.6 to +10.5 points, about +7 overall); 2 of 5 bands individually significant (only 118–259 patients at 15+ km per band).	PASS: not redundant	Effect is real but modest. Keep the "15+ km threshold" framing.
Distance model weight (new)	Logistic Regression AUC with vs without distance (5-fold CV)	Small gain if the effect is modest	0.6732 without, 0.6772 with distance (+0.004, inside the ± 0.01 fold-to-fold variation)	PASS	Explains the low model weight: small effect, small share of patients. No change to recommendation.
Double-risk segment	Reproduce ~4.7% of bookings / ~73% risk using lead ≥ 46 days & bracket ≥ 2	Match Week 6 figures	Got 2.46% / 82.1%, did not match	FAIL → FIXED	Root cause: correct definition uses lead ≥ 31 days, not ≥ 46 . Corrected definition reproduces 4.66% / 73.0% exactly. The Week 6 Decision Support text also cites 46+ days; superseded.
Double-risk segment	Bootstrap stability check on corrected segment (n=233)	Reasonably tight CI	95% CI [67.4%, 79.0%], SE $\approx$ 2.9 pp	PASS	Segment is stable; 51% relative lift over the 48.5% base rate

Component Tested	Test / Scenario	Expected Result	Actual Result	Pass/Fail	Action / Retest
DS collaborator's double-risk flag, Random Forest	5-fold CV AUC with ≥ 46 (as coded) vs ≥ 31 (validated)	No material difference if the model already learns the interaction	0.6661 ± 0.0060 vs 0.6662 ± 0.0056	PASS	Documentation fix, not a performance fix
DS collaborator's double-risk flag, exact Week 6 candidate (new)	Week 6 candidate rebuilt from its own code (Logistic Regression, 12 features, scaled, balanced classes); flag ≥ 46 vs ≥ 31 vs no flag; Week 6 CV method and 5×5 repeated CV	Threshold could matter more, since a linear model cannot learn the interaction alone	Candidate reproduced (0.6819 ± 0.0120 , matches candidate_model.json). Flag ≥ 46 : 0.6819 ; flag ≥ 31 : 0.6821 ; no flag: 0.6820 . 5×5 CV paired difference $+0.0001$ (31+ wins 13 of 25 folds).	PASS	No difference under either model, and the flag adds nothing once the interaction term is present. 31+ kept (see Section 6).

Note: one test genuinely failed on first attempt (the double-risk segment reproduction) and was root-caused and fixed. It is included here as real evidence rather than omitted, per the assignment's guidance to document actual testing outcomes rather than only successes.

The full testing record in the brief's required format (objective, issue identified, action taken, retest result, collaborating track and evidence for every test, including the two dashboard tests still pending) is in reports/Week7_Testing_Validation_Record.pdf and .csv.

4. Dashboard Testing Evidence

All four dashboard KPI tiles and charts were checked against their source CSVs and reproduced independently from the raw prepared dataset (Section 3, rows 1 to 5). No discrepancies were found between the dashboard, the bootstrap CI tables, and an independent recomputation. This confirms the Week 6 dashboard build is accurate and not dependent on any notebook-only logic. KPI 5 is shown on the dashboard as a lead-time-band chart without error bars. It was recomputed from the raw dataset and matched to the Week 6 KPI Validation report (Section 3, row 6); the chart's bars are read against those values during the dashboard test.

Filter/slicer testing (brief task 5): three dropdown slicers (appointment_type, age_group, reminder_sent) were added and tested against the expected values in reports/test_evidence/dashboard_filter_expected_values.csv. All four card values matched exactly for T0 (all data), T1 (Follow-up), T2 (age 65+) and T3 (reminder sent = Yes). **Status: PASS.** One observation: the confidence-interval error bars come from static lookup tables and do not update when a slicer is applied; bar values change correctly but the CI markers remain fixed. This is documented as a limitation (Section 8).

5. Distance Band Finding: Refinement Status

Testing (Section 3) reconfirmed that the three middle distance bands (<5km, 5–10km, 10–15km) are statistically indistinguishable, and that only the 15+km band is a genuinely distinct risk group. The Week 6 report text already reflects this downgrade, but the dashboard chart itself still displays four bars of steadily increasing height, which visually reads as a gradual trend rather than "three similar bands plus one distinct outlier."

New evidence this week supports that framing. Distance and lead time are unrelated (correlation -0.008), and the 15+ km gap shows up inside every lead-time band at about +7 points overall. So the 15+ km effect is real and is not an artefact of lead time, but it is modest, which is why the model gives it little weight.

A refinement was designed and validated in mockup form (outputs/dashboard/distance_chart_refinement_mockup.png): grouping the three overlapping bands under a shared muted color, isolating the 15+km band in a contrasting color, adding a bracket annotation reading "CIs overlap, not statistically distinct," and updating the chart title to state the finding directly.

Status: this refinement has been designed and specified (conditional formatting logic, annotation text, and exact Power BI implementation steps) but has **not yet been applied** to the live .pbix dashboard. It is logged as an open item ahead of final submission. See Section 8.

6. HC-POD Cross-Track Testing: Data Analytics ↔ Data Science

Track collaborated with	DS track collaborator (separate track; their findings informed this analysis). This was a real artefact-to-artefact check, not just a conversation.
Dependency	The DS track collaborator's double_risk_flag feature (train_improved.py) claims to encode the DA-validated double-risk finding (long lead time + 2+ prior no-shows → ~73% risk, ~4.7% of bookings)
Component tested	Whether the DS feature engineering actually reproduces the DA finding it is documented as representing
Information received from the DS track collaborator	Feature code: double_risk_flag = (booking_lead_days >= 46) & (previous_no_shows >= 2); code comment explicitly cites the DA 73%/4.7% finding as its rationale
Information provided to the DS track collaborator	The DA-validated segment definition and figures (lead ≥31 days & prior no-show bracket ≥2 → 4.66% of bookings, 73.0% no-show rate), independently reproduced from the prepared dataset. Handoff note and evidence package: handoff/ds_track_collaborator/ (delivery record completed when sent).
Testing activity	Recomputed the segment both ways (coded threshold vs DA-validated threshold) directly from the shared prepared dataset, then retested model performance under both the Random Forest and the exact Week 6 candidate (a Logistic Regression)
Finding	The coded threshold (≥46 days) is narrower and more extreme than the DA finding: 2.46% of bookings at 82.1% risk, vs the claimed 4.66% / 73.0%. It silently excludes 110 patients in the 31–45 day band who are part of the same validated stacking effect and still carry a 62.7% no-show rate.
Action taken	In a standalone retest (the Week 7 notebook), rebuilt the feature with the corrected threshold (≥31 days) and re-ran 5-fold cross-validation on the Random Forest with both versions. Then repeated the test on the exact Week 6 candidate model, rebuilt by importing the Week 6 feature code itself (Logistic Regression, 12 features, scaled, balanced classes), because that model type cannot learn the lead-time by prior-no-show interaction on its own, so the threshold could matter more there. Run with the Week 6 cross-validation method and with 5 × 5 repeated cross-validation, and with the flag removed entirely as a control. An earlier preliminary check on a simplified two-feature Logistic Regression reached the same conclusion; it is kept in the evidence folder and superseded by this test.
Retest result	Random Forest CV ROC-AUC: 0.6661 ± 0.0060 (original) vs 0.6662 ± 0.0056 (corrected); feature importance 0.0223 vs 0.0219. Exact Week 6 candidate (Logistic Regression), Week 6 method: 0.6819 ± 0.0120 (≥46, reproduces the Week 6 figure), 0.6821 ± 0.0120 (≥31), 0.6820 ± 0.0120 (no flag). 5 × 5 repeated CV paired difference +0.0001 (31+ wins 13 of 25 folds). No material difference in either model, and the flag adds nothing once the interaction term is present.
What changed as a result	No model performance change under either model. The flag covers only 2.5–4.7% of bookings, so it cannot move overall AUC; it works as a business rule for deciding who gets a call. Decision: keep the ≥31 definition. It reaches 233 patients at a 73.0% no-show rate, against 123 patients at 82.1% for ≥46, nearly twice as many high-risk patients at a still very high rate. Counts are on all 5,000 bookings; on the 4,737 attended and no-show bookings the model is trained on (cancelled excluded) they are 225 at 75.6% against 120 at 84.2%. This is a documentation/consistency fix, not a performance fix: the coded threshold should be corrected to ≥31 so code, comment and DA source figures agree. The Week 6 Decision Support text also states 46+ days; that wording is superseded by this correction.
Evidence	Segment size/rate comparison (both definitions); 5-fold CV ROC-AUC before/after and feature importance (Random Forest); exact-candidate Logistic Regression results (Week 6 CV method and 5 × 5 repeated CV). All computed directly from healthconnect_appointments_prepared.csv. Files: reports/cross_track_evidence/, notebook Sections 5 and 5b, scripts/week7_double_risk_exact_candidate_test.py with its output log and CSV, and handoff/ds_track_collaborator/ (note, segment comparison, retest CSVs, delivery record).
How this improved the overall solution	Removed a real inconsistency between what the DS model documentation claims and what the code does, verified it does not mask a larger problem (model performance is unaffected under either model), and gave the DS track collaborator a corrected, evidence-backed feature definition ahead of Week 8 final integration.

7. Updated Business Insights & Recommendations

- Booking lead time remains the strongest, most actionable predictor of no-show risk, validated by three independent methods across two tracks. Recommend this be the primary lever for any intervention (e.g. earlier reminder cadence for long-lead-time bookings).
- The double-risk segment (lead time ≥31 days AND 2+ prior no-shows) is a small but very high-risk group: 4.66% of bookings (233 of 5,000), 73% no-show rate, stability-confirmed via bootstrap. Definition kept at ≥31 after the exact-candidate Logistic Regression retest. It is a strong candidate for a targeted intervention pilot (e.g. a phone call rather than an SMS reminder); a design is in Section 9.

- Distance to clinic should not be treated as a graded risk factor. Only patients 15km+ away show a distinct risk profile, and this week's testing shows that gap is real inside every lead-time band (about +7 points) but modest. Messaging and resourcing should target that specific group rather than assuming a smooth distance gradient.
- The cancellation-to-no-show ratio falls from about 0.20 at 0–7 days to about 0.08 at 46–60 days, with non-overlapping intervals at the two ends. Long-lead-time bookings fail mostly through silent no-shows, not cancellations, which suggests pairing long-lead reminders with an easy reschedule option. This is a hypothesis to test, not a validated result.
- The reminder_sent effect is modest (51.4% vs 47.4% no-show rate) and should be interpreted cautiously. It was not a focus of this week's deeper testing and may warrant its own dedicated validation pass.

8. Limitations & Remaining Issues

Resolved this week

- Double-risk segment definition discrepancy between the DA report text and its initial, incorrect reproduction. Resolved: root cause identified, correct definition (≥ 31 days) documented and kept.
- DS track collaborator's double_risk_flag threshold mismatch against the DA-validated finding. Tested under two model types, quantified, and a corrected definition proposed and validated via retest.
- KPI 5 (cancellation-to-no-show ratio) coverage gap: reproduced and matched to Week 6.
- Week 6 open question on distance vs lead-time redundancy: answered. Not redundant (correlation -0.008); the effect is real but modest.

Unresolved / carried into Week 8

- Distance-band chart visual refinement is designed but not yet implemented in the live Power BI dashboard, so the "after" screenshot does not exist yet.
- The DS track collaborator's double_risk_flag has not yet been corrected in the actual train_improved.py codebase; the fix has been validated in a standalone retest but not merged back into their pipeline.
- reminder_sent and reminder_channel effects were not subjected to the same depth of segment testing this week and remain a lower-confidence finding relative to lead-time and distance.
- The double-risk call pilot has been designed (Section 9) but not run. It needs a real intervention window and clinic agreement.
- Week 6 item: re-run the KPI 1 bootstrap for the "3+" prior no-show bracket if more data becomes available. No new data was available this week, so it was not re-run.
- Week 6 item: independent review of the DS feature-selection table and robustness check by a DS or ML Engineering reviewer has not been done; the cross-track test covered only double_risk_flag.
- The in-band distance test compares under-15 km with 15+ km only, uses normal-approximation intervals rather than a bootstrap, and excludes 90 rows with unknown distance. Individual band results are noisy because of small 15+ km groups.

New issue discovered during testing

- Documentation-to-code mismatches (like the double_risk_flag threshold) are a risk pattern worth checking systematically across both tracks before Week 8 final integration. The same pattern appeared in the Week 6 Decision Support text, which states the flag as 46+ days. That wording is superseded by the ≥ 31 definition validated here.

Limitations, potential impact and recommended mitigation

Limitation	Type	Potential impact	Recommended mitigation	Before Week 8?
Single dataset of 5,000 fictional bookings; no held-out or external data. All model checks are cross-validation on the same rows.	Data / Testing	Findings and AUC values may not carry over to new data; AUC could be optimistic.	Describe results as validated on this dataset only; add a held-out or new-data check before final integration.	Recommended

Limitation	Type	Potential impact	Recommended mitigation	Before Week 8?
90 of 5,000 rows have no distance and are excluded from distance tests.	Data	Small (1.8%); unlikely to change the direction of the distance findings.	Documented; run a sensitivity check if distance is used in the final model.	No
The 3+ prior no-show bracket is small with a wide interval (Week 6 item).	Data / Analytical	KPI 1 top bracket is imprecise.	Keep flagged as fragile; re-run the bootstrap if more data arrives.	No
Distance groups inside lead-time bands are small (118 to 259 patients at 15+ km); simple intervals used; distance not tested against prior no-shows.	Analytical	Single-band results are noisy; redundancy with prior no-shows is not ruled out.	Report the direction, not per-band significance; bootstrap and add the prior no-show check in Week 8.	Optional
The flag test ran on the 4,737 attended and no-show bookings the Week 6 model uses (cancelled rows excluded), so its counts differ from the DA figures on all 5,000 bookings (225 vs 233 patients, 120 vs 123).	Modelling / Documentation	Two sets of counts for the same segment can confuse a reader.	State the denominator every time: DA figures on 5,000 bookings, model figures on 4,737.	No
Independent review of the DS feature-selection table and robustness check (Week 6 item) not done.	Modelling / Integration	Self-checked results remain unreviewed.	Ask the DS track collaborator or an ML Engineering reviewer.	Recommended
Distance chart refinement not applied in the live .pbix; dashboard slicers did not exist and are not yet tested.	Testing	A misleading chart remains and filtered behaviour is unverified.	Follow the Power BI steps, export the after screenshot, record slicer results.	Yes
Confidence-interval error bars come from static tables computed on the full dataset.	Technical	May not respond to slicers, which could mislead a filtered view.	Observe during slicer testing; document it or exclude those charts from slicer interactions.	Yes
Corrected double_risk_flag threshold not yet merged in the DS collaborator's code.	Integration	Code, comment and figures disagree until merged.	Send the handoff package; re-run the comparison on the merged pipeline.	Yes
Documentation-to-code mismatches (also present in the Week 6 text).	Process	Other hidden mismatches may exist across tracks.	Run a systematic doc-versus-code check before final integration.	Recommended
Confidence-interval error bars are computed on the full dataset and do not respond to dashboard slicers.	Technical / Visual	A filtered view shows correct bar values but static CIs, which could mislead a reader comparing filtered subsets.	Add a note to the dashboard or exclude CI charts from filtered views; review in Week 8.	Yes
Distance-band bar colour differentiation was not applied in the live dashboard. All four bars share the same colour.	Visual	The chart may still read as a gradual trend to a casual reader despite the CI-overlap annotation.	Apply conditional bar colour in Power BI in Week 8 (steps in the guide) and re-export.	Yes
The call pilot has not been run.	Project	No real-world evidence yet that the intervention lowers no-shows.	Agree the design with clinic staff; run it, or present it as planned work.	Decision needed

New cross-track dependencies: Week 8 final integration depends on the DS track collaborator merging the corrected threshold, and the Data Analytics re-run on the merged pipeline depends on that merge.

9. Double-Risk Call Pilot: Design (for Week 8)

Week 6 asked for a pilot of the double-risk intervention. It cannot be run inside this dataset, so this is a design only. Items marked proposed need agreement with clinic staff.

Element	Design
Who is flagged	Bookings with lead time ≥ 31 days AND 2+ prior no-shows: about 4.66% of bookings (233 in this 5,000-booking dataset), baseline no-show rate 73.0%

Element	Design
Intervention (proposed)	A staff phone call to the patient in place of the standard SMS reminder
Comparison	Randomly split flagged patients into call vs standard reminder, same clinic and same period, so the two groups are comparable
Primary metric	No-show rate. Secondary: cancellation/reschedule rate (the KPI 5 ratio) and staff minutes per call
Planning estimate	To detect a 10-point drop (73% to 63%) at 5% significance and 80% power needs about 341 flagged patients per group (682 total), which is roughly 14,600 bookings at the 4.66% flag rate. A smaller effect needs more, so several months of bookings, or several clinics, will be needed.
Success rule (proposed)	A drop of at least 10 points, with the 95% interval excluding zero
Status	Design only. Nothing has been run and no pilot results are claimed.

10. Validated Findings That Should Influence the Wider HealthConnect Solution

This section is what the Data Analytics track hands to the other tracks and to Project Management (brief tasks 14 and 15).

Validated finding	Evidence	Who should use it	How	Confidence
Lead time is the strongest predictor: no-show rate rises from 27.8% (0–7 days) to 67.7% (46–60 days).	Week 5 estimates, Week 6 bootstrap CIs, Week 7 segment tests (appointment type, age group), DS model importance.	DS, ML Engineering, Project Management	Keep booking_lead_days as a core model feature and a required, range-checked pipeline input (the data covers 0 to 60 days). Prioritise long-lead bookings for any reminder or intervention work.	High
Double-risk segment: lead time ≥31 days and 2+ prior no-shows, 4.66% of bookings, 73.0% no-show rate.	Reproduced from the data; bootstrap 95% CI [67.4%, 79.0%]; threshold retest under two model types.	DS (feature definition), Project Management (pilot)	Use ≥31 in double_risk_flag. Use this group as the pilot target (Section 9).	High for the association, not proof of cause
Distance is a 15+ km threshold effect, real but modest (about +7 points, present in all 5 lead-time bands).	CI overlap check; in-band comparison; correlation –0.008 with lead time; AUC gain +0.004.	DS, Project Management	Model distance as a 15+ km indicator, not a smooth gradient. Do not resource a graded distance programme.	Medium
Cancellations shrink relative to no-shows as lead time grows (ratio about 0.20 down to 0.08).	KPI 5 reproduction; intervals at the two ends do not overlap.	Project Management, GenAI (reminder wording)	Hypothesis to test: pair long-lead reminders with an easy reschedule or cancel option.	Medium (finding), low (the action)
Reminder effect is modest (51.4% vs 47.4% no-show rates).	KPI 4 reconciliation only; not deeply tested.	Project Management	Do not rely on reminders alone. Validate reminder_sent and reminder_channel before recommending changes.	Low to medium

11. Week 8 Recommendations

#	Recommendation	Owner	Reason
1	Apply the distance-chart refinement in the live dashboard and export the after screenshot.	Data Analytics	Required refinement and retest evidence; the chart currently implies a gradual trend.
2	Add slicers and test them against the expected values in reports/test_evidence/dashboard_filter_expected_values.csv; record the results.	Data Analytics	Brief task 5 (filters) is untested.
3	Merge ≥ 31 into double_risk_flag in train_improved.py and update the comment; re-run the flag comparison on the merged pipeline.	DS track collaborator, Data Analytics	Removes the code-versus-documentation mismatch.
4	Add a held-out or new-data check of the flag threshold on the exact Week 6 candidate, and fold the result into the final model comparison.	Data Analytics, DS track collaborator	The exact-candidate retest is cross-validated on one dataset only.
5	Agree the call pilot design (intervention, success rule, sample size) with Project Management and clinic staff.	Project Management, Data Analytics	Week 6 asked for a pilot; it is designed but not run.
6	Test reminder_sent and reminder_channel to the same depth as lead time and distance.	Data Analytics	Reminder findings are the least tested.
7	Run a systematic documentation-versus-code check across tracks before final integration.	All tracks, coordinated by Project Management	The 46 versus 31 mismatch was found by chance.
8	Use the validated findings (lead time, double-risk ≥ 31 , 15+ km threshold, KPI 5 pattern) as the storyline for the final presentation.	Data Analytics	Keeps the final story limited to what has been tested.